

## POTPRESS · REQUIREMENTS

# PotPress requirements

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## DOCUMENT CONTROL

|             |                                       |         |                |
|-------------|---------------------------------------|---------|----------------|
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| Status      | <b>Draft</b>                          | Date    | 2026-09-25     |
| Author      | Amish Chadha                          | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept         |         |                |
| Repository  | github.com/BoujeeEnjinia1701/potpress |         |                |
| Source      | docs/03-requirements.md @ 41dbb03     |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                                                                           |
|---------|------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.4     | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002); R3 at 10 t with a metal stop, R9 pin interlock, R10 \$930, R11 rack 0.8 x 0.8 m and folded rails, R12 lead-free scrap; status from PPR-CAL-001 v0.2 |
| 0.3     | 2026-09-25 | Amish Chadha | Apply Amish's 2026-09-25 decisions (PPR-DDR-001); R10 target \$720, R6 band decided; status from PPR-CAL-001; proposed changes to R3 and R11                                                     |
| 0.2     | 2026-09-25 | Amish Chadha | First measurable requirements for TRL 2, with concept status                                                                                                                                     |
| 0.1     | 2026-09-24 | Amish Chadha | Initial scaffold                                                                                                                                                                                 |

## PotPress requirements

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These requirements were checked by calculation at TRL 3 (PPR-CAL-001 v0.2). They are not yet validated with a filter factory and will be revised after co-design sessions (see PPR-PRB-001). One is **not met** on paper (R10 cost), three are at risk (R2, R7 and R9) and eight are met on paper; see Table 2. Amish's decisions of 2026-09-25 are recorded in PPR-DDR-001 and PPR-DDR-002: the R6 default band, R3 judged at the 10 t working force, the R10 target of \$930, the R11 rack area of 0.8 x 0.8 m, the pin-presence interlock in R9 and lead-free scrap in R12.

The **reference filter** used throughout is the common flowerpot form: inner rim diameter 280 mm, inner depth 240 mm, wall 15 mm, flat rim about 345 mm across, about 12 L to the brim and about 10 L working volume (dimensions are estimates based on the 280 by 250 mm form reported by [Potters for Peace](#)).

*Table 1. Requirements.*

| ID  | Requirement                       | Target                                                                                                                                                                                                                                                                                                 | Verification (TRL 3 or later)                          |
|-----|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| R1  | Form the reference filter         | Inner rim 280 mm, inner depth 240 mm, wall 15 mm, flat rim; geometry set by parameters in one source file                                                                                                                                                                                              | Model check; later measurement of pressed pots         |
| R2  | Even wall                         | Wall thickness within $\pm 1$ mm around the circumference and from pot to pot; molds self-center to within 0.5 mm                                                                                                                                                                                      | Tolerance stack and guide design; later sectioned pots |
| R3  | Pressing force and frame strength | 20 t (196 kN) rated jack; frame, pin, bolts and platen carry 1.5 times the jack rating (294 kN) without yield; deflection between the molds under 1 mm at the 10 t maximum working force, with the molds closing on a metal stop so frame stretch does not set the wall (decided by Amish, 2026-09-25) | Beam and pin calculation; later proof load             |
| R4  | Throughput                        | Press cycle of 6 min or less with two operators, so 50 or more pots per 6 h of pressing                                                                                                                                                                                                                | Cycle breakdown; later timed trial                     |
| R5  | Mold opening                      | Male mold lifts at least 30 mm clear of the female rim so the mold carriage slides out for loading and demolding; pressing stroke within one jack stroke                                                                                                                                               | Kinematic layout                                       |
| R6  | Flow-rate QC                      | Rack tests 4 soaked filters at once; T-gauge reads flow over 1 h to 0.1 L/h; repeatability $\pm 0.1$ L/h; acceptance band set per factory (default 1.0 to 2.5 L/h in the first hour, corrected to 25 °C; decided by Amish, 2026-09-25)                                                                 | Gauge calibration by volume; later repeat tests        |
| R7  | Molds made locally                | Molds cast from 3D-printed patterns printed on a 250 x 250 mm class printer; finishing with hand tools and a drill press; no lathe over 300 mm swing                                                                                                                                                   | Pattern split and finishing review with a foundry      |
| R8  | Frame built locally               | Standard steel channel and plate; stick welding, drilling and bolting only; heaviest single part 40 kg or less for two-person handling                                                                                                                                                                 | Part list and mass estimate                            |
| R9  | Safe operation                    | Fixed guards on the sides and back, interlocked front gate; jack pump and release outside the guard; a pin-presence interlock prevents pressing unless the load pin is fully home (decided by Amish, 2026-09-25)                                                                                       | Hazard review; later guard check                       |
| R10 | Affordable                        | Press plus QC rack \$930 or less in parts (raised from \$600 to \$720 and then to \$930 by Amish, 2026-09-25)                                                                                                                                                                                          | Priced BOM ( bom/ bom.csv )                            |
| R11 | Footprint                         | Press within 1.0 x 0.7 m floor area and 2.0 m height, with the rail extension folded; QC rack within 0.8 x 0.8 m (relaxed from 0.8 x 0.5 m by Amish, 2026-09-25)                                                                                                                                       | Model check                                            |
| R12 | Product-safe materials            | Faces that touch clay or test water are aluminum, food-grade polyethylene or stainless steel; no lead-based paint or oiled release agents on mold faces; molds cast from lead-free scrap (no free-machining alloys; decided by Amish, 2026-09-25)                                                      | Material list review                                   |

Table 2. Status against each requirement at TRL 3 (from PPR-CAL-001 v0.2, not met first).

| ID  | TRL 3 value                                                                                                                                                             | Status                                              |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| R10 | Priced BOM \$984 (press \$903, QC rack \$81)                                                                                                                            | <b>Not met</b> , \$54 (6 %) over                    |
| R2  | Wall $\pm 1.40$ mm as cast; $\pm 0.43$ mm with molds hand-finished to templates; coaxial $\pm 0.43$ mm via the locating lip                                             | <b>At risk</b>                                      |
| R7  | 16 pattern segments for a 250 mm printer; cavity finishing without a lathe unproven                                                                                     | <b>At risk</b>                                      |
| R9  | Guards, gate, jack-release interlock and pin-presence interlock in the BOM, not modeled                                                                                 | <b>At risk</b>                                      |
| R1  | Reference filter from <code>PARAMS</code> in <code>cad/src/model.py</code> : 12.30 L to the brim, 9.91 L working                                                        | Met on paper                                        |
| R3  | No yield at 294 kN: beams 190 MPa, uprights 54 MPa, 60 mm pin 258 MPa (yield 650), joint bolts 75 MPa. Deflection between the molds 0.61 mm at 10 t (1.84 mm at 294 kN) | Met on paper                                        |
| R4  | 5.2 min cycle, 69 pots per 6 h                                                                                                                                          | Met on paper, thin margin                           |
| R5  | 310 mm opening against 270 mm needed; 110 of 150 mm jack stroke                                                                                                         | Met on paper                                        |
| R6  | 4 stations; 0.1 L is 1.69 mm on the gauge; temperature correction to 25 °C                                                                                              | Met on paper; repeatability not verifiable at TRL 3 |
| R8  | Heaviest part 38.9 kg (platen); upright joints bolted, largest frame part 37.4 kg                                                                                       | Met on paper                                        |
| R11 | Press 840 x 655 mm with the rail extension folded (970 mm deployed), 1,806 mm tall; QC rack 780 x 780 mm                                                                | Met on paper                                        |
| R12 | Aluminum faces, polyethylene liners, HDPE buckets; lead-free scrap alloy                                                                                                | Met on paper                                        |

### Requirement changes decided by Amish, 2026-09-25 (PPR-DDR-002)

- **R3:** deflection judged at the 10 t maximum working force, with the molds closing on a metal stop (was 294 kN; status not met to met on paper).
- **R10:** target raised from \$720 to \$930. The decided bolted joints and hinged rail extension bring the BOM to \$984, so R10 is still not met; the options for the remaining \$54 are proposed, awaiting Amish (PPR-DDR-002).
- **R11:** QC rack area relaxed from 0.8 x 0.5 m to 0.8 x 0.8 m; the press footprint is judged with the hinged rail extension folded (status not met to met on paper).
- **R9 and R12:** the pin-presence interlock and lead-free scrap are now part of the requirement text.

### ■ Assumptions

- Pressing force needed for a well-consolidated wall is not published in the sources found (one low-cost press forms round-bottom filters with a 2 t jack, [Henry, Maley and Mehta, 2013](#)); 5 to 10 t is assumed as the working range and the 20 t jack rating is used as the design load (estimate, to be confirmed with a partner factory).
- Mix follows the RDI-C recipe reported by [Rayner \(2009\)](#); the reference filter needs about 7.5 kg of mix per pot, below the 8 to 9.5 kg factories report (PPR-CAL-001).
- The default flow acceptance band of 1.0 to 2.5 L/h follows Nicaraguan practice and the Potters for Peace range of 1.5 to 2.5 L/h; each factory sets its own band.
- Allowable stress in structural steel is taken as 165 MPa (S275 with a factor of about 1.67).